

Causal Inference 2025 - Final Project

Analysis of Randomized Studies with Noncompliance

This assignment requires the use of R. Do not include code in your written report: instead, provide a clear and concise presentation of your results, avoiding copy-pasted R output.

The report may be written in Italian or English and must be submitted in PDF format. Partial credit based on your code may be awarded for some questions.

- Please upload both your report and the R code file to Moodle at least two days before your oral exam date. For example, if your oral exam is scheduled for June 11, you must upload the files by 11:59 PM on June 8.
- Use the following file naming convention:

SurnameName.pdf and SurnameNameRCode.R,

For the oral exam you need to prepare a presentation, based on the report.

Datasets

Select one of the following datasets (available on Moodle).

The Faenza Study: The dataset `BSE.txt` contains data on the Faenza Study, a randomized experiment that compares two methods of teaching Breast Self-Examination (BSE) with the aim of assessing whether teaching programs could increase BSE practice and improve examination skills. The two methods consist of either email information about BSE (standard treatment) or email information and the attendance of a course (new treatment) involving theoretical and practical sessions.

The dataset includes the following variables:

- Covariates: `ypr` - an indicator for previous BSE practice, `know1` - an indicator of good knowledge of breast pathophysiology; and `age` - age in years.
- Assignment variable: `z` (1 for assignment to the new treatment, 0 for assignment to the standard treatment);
- Treatment receipt indicator: `w` (1 for new treatment, 0 for standard treatment);
- Outcome variable: `y` - binary indicator for whether BSE was practised 1 year after the course.

The Influenza Study: The dataset in `flu.txt` is from a randomized encouragement design of McDonald et al. (1992), which was also re-analyzed by Hirano et al. (2000). The study, conducted between 1978 and 1980 at a general medicine clinic in Indiana (USA), aims to assess the effect of receiving the influenza vaccine on hospitalization for flu-related illnesses during the winter. Physicians were randomly selected to receive a letter encouraging them to inoculate patients at risk for flu. The units of analysis are patients. The treatment of primary interest is each patient’s actual receipt or not of the influenza vaccine, not the randomly assigned encouragement in the form of the letter sent to each patient’s physician.

The dataset includes the following variables:

- Covariates: **age** - age measured in years, **copd** - a binary indicator for chronic obstructive pulmonary disease) and **heart** - an indicator for prior heart problems;
- Assignment variable: **z** (1 if a letter was sent to the physician, and 0 otherwise);
- Treatment receipt indicator: **w** (1 for patients who received the vaccine and 0 otherwise);
- Outcome variable: **y** - binary indicator for hospitalisation for flu-related illnesses.

The JOBSII Study: The Job Search Intervention Study (JOBS II) was a randomized field experiment that investigated the efficacy of a job training intervention on unemployed workers.

The dataset, `JobsII.txt` includes the following variables:

- Covariates: **age** - age measured in years, **sex** - Indicator variable for sex (1 = female), **married** - Indicator variable for married (1 = Married), **nonwhite** - Indicator variable for race (1 = nonwhite). **educ** - Indicator variable for higher education (1 = Some college or more), **income** - Factor with five categories for level of income (You need to create dummies for use this variable in models), **depress_pre** - Measure of depressive symptoms pre-treatment.
- Assignment variable: **z** - Indicator variable for whether participant was randomly selected for the JOBS II training program (1 = assignment to participation)
- Treatment receipt indicator: **w** - Indicator variable for whether participant actually participated in the JOBS II program (1 = participation.)
- Outcome variables (Select one outcome variable): **depress** - Measure of depressive symptoms post-treatment, **emp** - Indicator variable for employment (1 = employed), **job_seek** - A continuous scale measuring the level of job-search self-efficacy with values from 1 to 5.

Outline for the report and presentation

1. Describe the noncompliance problem, specifying if you have one-sided or two-sided noncompliance.
2. Report the mean of each variable for the full sample, as well as for the subsamples classified by assignment status and treatment status.
3. Examine the means of the pre-treatment variables. Can we assume that treatment assignment is unconfounded given the observed covariates? Can we also assume that the actual receipt of the treatment is unconfounded given the observed covariates? Briefly explain your reasoning. Why are these questions important?
4. Estimate the overall intention-to-treat effect on the primary outcome, Y , with its standard error. What is the meaning of this estimate in your study?
5. Define the compliance types: Compliers, Never-takers, Always-Takers and Defiers (if they exist)
6. Briefly discuss the monotonicity assumption in your study
7. Exclusion restrictions.
 - (a) Exclusion restriction for never-takers: Briefly discuss a reason that we might be suspicious of the exclusion restriction for never-takers, in the study. How far-fetched are your suspicions about the exclusion restriction? Do you think it is valid to assume the exclusion restriction for never-takers in your study?
 - (b) Exclusion restriction for always-takers (if there exist): Briefly discuss a reason that we might be suspicious of the exclusion restriction for always-takers, in the study. How far-fetched are your suspicions about the exclusion restrictions? Do you think it is valid to assume the exclusion restriction for never-takers in your study?
8. Assume monotonicity and exclusion restrictions. Estimate the complier average causal effect on Y , using the Wald estimator (IV or ratio of ITTs). Calculate the standard error of the estimator using the Delta method. What is the meaning of this estimate in your context?
9. Compare the complier average causal effect (CACE) estimate with the overall intention-to-treat (ITT) estimate. From a policy perspective, which estimate is more informative or valuable? From the perspective of an individual deciding whether to take the treatment, which estimate is more relevant? Briefly discuss.
10. Assume monotonicity and exclusion restrictions. Derive the proportion of each compliance status, the mean of the outcome for never-takers and always-takers and the mean of the outcome for compliers under assignment to treatment and under assignment to control. Compute the method-of-moment estimator of this quantities and of CACE. Calculate the standard error of the estimators using the bootstrap.

11. In general, why is the method-of-moments estimator for the complier average causal effect not ideal?
12. Bayesian model-based approach under monotonicity and exclusion restrictions.
 - (a) Write down the *observed likelihood function*, i.e. the likelihood function for the observed data, without knowing the compliance status for each unit.
 - (b) Write down the *complete likelihood function*, i.e. the likelihood function, had we also known the compliance status for each unit.
 - (c) Assuming flat priors for all parameters, use a Gibbs sampler to derive the posterior distribution of parameters. Before you start the sampler, take initial values from Beta(1,1)/Dirichlet(1,1,1) distribution for the compliance status probabilities. Run the chain for 2000 cycles, and based on the last 1000 cycles of the chain, report the mean, standard deviation and 0.025 and 0.975 quantiles of the posterior distribution of (i) the proportion of each compliance status, (ii) the mean of the outcome for never-takers and always-takers; (iii) and the mean of the outcome for compliers under assignment to treatment and under assignment to control; (iv) ITT effect; (v) CACE. Comment on the results.
 - (d) Repeat points (a)–(c) while relaxing the exclusion restriction assumptions for never-takers, always-takers, or both. Report the results and discuss whether the exclusion restrictions appear plausible. **Hint:** Plot the posterior distribution of the ITT for never-takers and/or always-takers, as well as the scatter plot of the posterior draws of the ITT for compliers against the ITT for never-takers and/or always-takers.
13. Conclude with a discussion based on the results you obtained.